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Development of Java Island-scale Flood Model using CaMa-Flood

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Abstract. Java Island is one of the most densely populated islands in Indonesia, with a total population of 152 million. The continuous population increase has led to a rise in disaster vulnerability, including flood disasters. Since 2012, the frequency of floods has continuously increased, with an average annual rise of 13.18%. Global-scale-river models were developed to evaluate the effects of runoff forcings on food hazard, droughts, and water resources management. Indonesia, including Java Island, has limitations in data storage and management, resulting in some areas having inadequate data for flood modelling. The Java Island-scaleflood-model was developed using satellite-based runoff data as the primary input. The model enabled historical flood reconstruction on the entire Java Island, which was chosen due to the frequent and increasing flood incidents. The modelling system uses the Catchment-based Macro-scale Floodplain (CaMa-Flood) version 4.2.0 river hydrodynamics model, which simulates hydrodynamic processes at a resolution of approximately 1 km. The CaMa-Flood simulation is driven using daily simulated runoff from the Earth2Observe (E2O) model, a global river map from MERIT Hydro, and a global topography map from MERIT DEM. The results of this model are the conditions and historical data of river discharge, and inundation area on the island of Java. The use of global-scale river models has not yet been widely implemented in Indonesia. Therefore, this research is conducted to understand changes in flooding on an island and will advance climate system studies and assist both non-governmental and governmental institutions in formulating flood mitigation policies for the Java Islands.

1. Introduction

Climate change has become one of the major threats to the hydrological cycle. Recent studies show that climate change is causing an increase and alteration in temperature and precipitation, causing a threat of extreme weather events such as the risk of flooding worldwide[1]. Flooding is

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a common natural phenomenon that occurs due to a rise in water levels, leading to inundation in certain areas [1]. The massive population growth has led to an increase in high surface runoff and tidal surges, which are the factors that contribute to floods. The impacts of flood disasters include economic losses, casualties, and material damage, particularly in developing countries. Indonesia, classified as a developing country, experiences frequent flood events. Java Island, one of the islands in Indonesia, is one of the most densely populated areas [1]. The continuous population increase has led to a rise in disaster vulnerability, including flood disasters. According to data from the Indonesian National Disaster Management Agency, Java Island experienced 560 flood events in 2021, accounting for 48.25% of all flood disasters in Indonesia, with a concerning trend of an average annual increase of 13.18% since 2012 [1]. This indicates significant challenges related to climate change, urbanization, and flood disasters on Java Island [2].

Global-scale River models were developed to evaluate the effects of runoff forcings on flood hazard, drought, and water resources management [3]. These models have developed in recent years and are commonly used in various studies, including large-scale flood modeling and the impacts of climate change on floods [4]. The Java scale flood model was developed using satellitebased runoff data as the primary input. The global-scale model utilizes a river routing model to calculate river discharge and inundation area, which then could be regionalized. The model enabled historical flood reconstruction on Java Island, which was chosen due to frequent and increasing flood incidents. This study was conducted to model flood hazards and risks on a large to global scale using the Catchment-based Macro-scale Floodplain (CaMa-Flood) model, a large- to global-scale hydrodynamic model due to its ability to realistically simulate river and floodplain hydrodynamics, which simulates hydrodynamic processes at a resolution of approximately 1 km [5]. CaMa-Flood uses the local inertial approximation to solve the mass and momentum equations for river and floodplain flows [6]. The CaMa-Flood simulation is driven using daily simulated runoff from the Earth2Observe (E2O) model with a resolution of 0.25 degrees, a global river map from MERIT Hydro, and a global topography map from MERIT-DEM [5].

The results of this model are the conditions and historical data of river discharge and inundation areas on the island of Java. This research is conducted to understand changes in flooding on a global scale, particularly in Java Island. Furthermore, the enhanced predictability offered by the proposed model will advance climate system studies and assist both nongovernmental and governmental institutions in formulating flood mitigation policies for the Java Islands.

2. Materials and Method

2.1 Study Area

Indonesia is located between Southeast Asia and Australia, with its main islands being Java, Bali, Sumatra, Kalimantan, Sulawesi, and Papua. The study area in this research is the entire island of Java, located in Southeast Asia, bordered by the Indian Ocean to the south and the Java Sea to the north. **Figure 1**, as seen on the red line, shows the location of Java Island in Indonesia, which lies within 105° 07’ 0.00” E to 114° 37’ 0.00” E longitude and 5° 48’ 0.00” S to 8° 37’ 0.00” S. Java Island is composed of six provinces, including Banten, Special Region of Jakarta, West Java, Central Java, Special Region of Yogyakarta, and East Java.

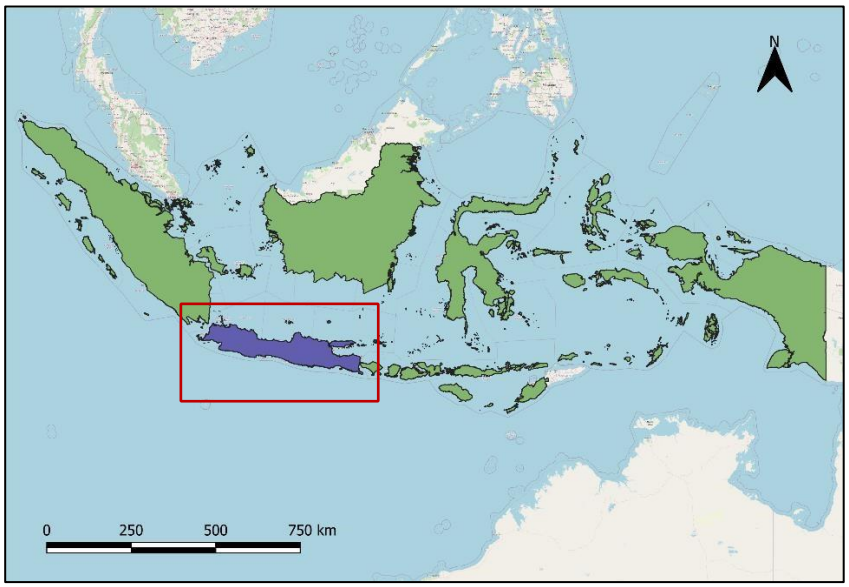


Figure 1. Location of Java Island in Indonesia. The red box indicates the extent of Java Island, and the green-shaded areas represent the area of Indonesia

Characterized by a tropical climate with high temperatures and humidity, Java Island experiences substantial rainfall throughout the year [7]. The flood season lasts from November to March, and the dry season lasts from April to October. The combination of high rainfall and rapid population growth has led to land use changes and increasing the risk of flooding on Java Island. The flood season refers to the period when the water level of a river rises seasonally due to seasonal precipitation in the watershed, which can easily cause flood disasters [7].

2.2 Runoff Data

CaMa-Flood uses runoff data as an input for simulations. The model simulation is driven using daily runoff simulated data from the Earth2Observe (E2O) water resources reanalysis (WRR2) runoff model, produced by the land surface hydrological model Hydrology Tiled ECMWFS Scheme for Surface Exchanges over Land (HTESSEL). The HTESSEL land surface model (LSM) calculates how the land surface responds to atmospheric conditions, providing estimates of surface water and energy fluxes, as well as the temporal dynamics of soil temperature, soil moisture, vegetation interception, and snowpack conditions [8]. The runoff data are distributed to each unit catchment according to the overall proportion of the unit catchment in the corresponding grid [9]. **Table 1** provides information related to the raw runoff model data used in hydrodynamic CaMa-Flood model.

Table 1. Eath2Observe Runoff Model Data

Data	Resolution	Source
Surface Incident Longwave Radiation (LWdown)	3-Hourly | Daily	ECMWF
Surface Atmospheric Pressure (PSurf)	3-Hourly | Daily	ECMWF
Near Surface Specific Humidity (QAir)	3-Hourly | Daily	ECMWF

Data	Resolution	Source
Rainfall Rate (Rainf)	3-Hourly | Daily	**MSWEP** (Multi-Source Weighted-Ensemble Precipitation)
Surface Incident Shortwave Radiation (SWdown)	3-Hourly | Daily	ECMWF
Air Temperature (Tair)	3-Hourly | Daily	ECMWF
Wind Speed (Wind)	3-Hourly | Daily	ECMWF
Land Surface Map (LSM)	3-Hourly | Daily	ECMWF

The runoff data have a spatial resolution of 0.25° x 0.25° with daily temporal scale. The aim of this study is to perform river routing model with the best model resolution (1 arcmin). Therefore, it is necessary to downscale the input parameters to achieve the same spatial resolution. The CaMa-Flood package provides a script for runoff input matrix interpolation in binary format to address this issue.

2.3 CaMa-Flood Model

CaMa-Flood is a global hydrodynamic model extensively used in global and regional studies to simulate the hydrodynamic behaviour of rivers and floodplains [5]. The model was designed based on the unit catchments approach, where each unit incorporates discretized topographical parameters for rivers and floodplains, allowing for a more accurate representation of sub-gridscale processes [10]. The model utilizes the local inertial equation, a simplified formulation of the one-dimensional Saint-Venant equation derived by neglecting the advection term [10]. This approach enhances computational efficiency when simulating river hydrodynamics on a continental scale. The model simulates river and floodplain hydrodynamics (such as river discharge, water surface elevation, flooded area, and surface water storage), routing runoff from a land surface model to oceans or inland seas through a predefined river network map [10]. This approach is highly effective for modelling flood depth and conducting a more detailed analysis of river flooding, particularly in regions with seasonal rainfall patterns, such as Java Island [5].

The model simulation is driven using topography data, global river width, and daily simulated data from the Earth2Observe (E2O) water resources reanalysis (WRR2) runoff model, as shown in **Table 2** [10].

Table 2. Data Source for CaMa-Flood Model

Data Name	Resolution	Sources
Global River Map	3 arc-second (~90 m)	MERIT Hydro
Global Topography Data	3 arc-second (~90 m)	MERIT DEM
Runoff Data	0.25° (~20 km)	Earth2Observe (E2O)

The results of streamflow are downscaled to a higher resolution of 1-arcmin (~ 1.8 km). Streamflow is generated by combining several datasets (topography maps, runoff data, and global river width). The CaMa-Flood package provides topography and global river data at various resolutions (15 min, 6 min, 5 min, 3 min, and 1 min) based on MERIT-DEM and GWD-LR. This study uses the default parameters of CaMa-Flood to broaden and enhance the understanding of the model's baseline performance. Additionally, it is expected to enable future studies to apply the model in new ways. In the downscaling methodology applied, flood depth at each pixel is

determined based on the flood depth at the outlet pixel and the relative elevation of each pixel. In the default downscaling scheme, only flooding from the main channel of each catchment unit is considered, potentially underestimating flooding from tributaries.

3. Result and Discussion

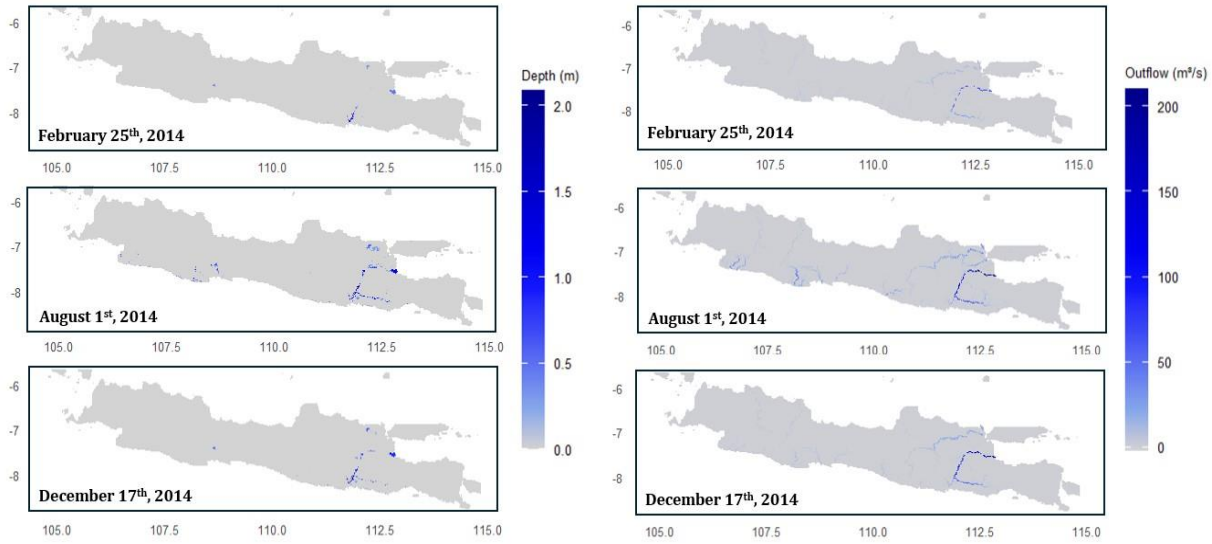


Figure 2. River Discharge Map along The Island of Java

The simulation period is set in 2014, considering the availability of the Earth2Observe (E2O) forcing dataset (1980-2014), allowing the model to capture the most recent conditions. Fig. 2 shows the simulated daily outflow at a ~ 1.8 km resolution for one year. The CaMa-Flood model can produce outputs in the form of river discharge and inundation areas across the island of Java. From the model simulation results, August is identified as a month with high rainfall and runoff, as evidenced by the clear depiction of the river network in that month. The maximal outflow formed along the island of Java is between 150-200 m^3/s, located in the Brantas River and the Bengawan Solo River. The first day (February 25th, 2014) is presented to show a relatively low level of flooding, with flooding of 1.0-1.5 m. However, flood depths are slightly higher in the middle of the year, with 1.5–2.0 m flooding occurring in Brantas, Bengawan Solo, and Citanduy River Basin.

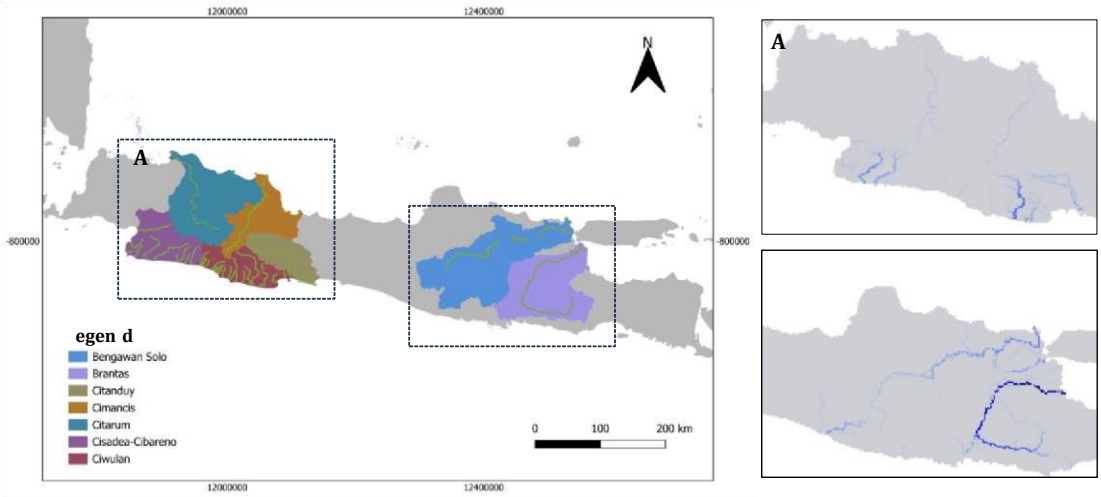


Figure 3. The Comparison Between River Network Map and The CaMa-Flood Model.
(A) Detailed of River Network at The West Java of Indonesia
(B) Detailed of River Network at The East and Central Java of Indonesia

Research shows that the Catchment-Based Macro-scale Floodplain (CaMa-Flood) model simulates the dynamics of floodplain inundation and river discharge along the primary channels of major river basins [11]. It is also shown in Java Island that the CaMa-Flood model can clearly represent the floodplain and river network along the main rivers of the basins, as seen in several river basins, including the Brantas, Bengawan Solo, Citanduy, Cimanuk-Cisanggarung, Citarum, Cisadea-Cibareno, and Ciwulan-Cilaku basins (**Figure 3**). The simulation results also show that the model can only identify the large main river networks, which are part of watersheds with a large area. The results of the model with CaMa-flood simulation indicate that reservoirs or lakes cannot be discretized, as shown in **Figure 3**. This is due to the fact that lakes and reservoirs typically require more specialized algorithms and data to calculate their storage and flow regulation processes[12].

4. Conclusion

The CaMa-Flood model applied to the island of Java has proven capable of accurately defining a river network on a large scale. This study shows a river's hydrologic and hydrodynamic changes in Java Island, providing continuous information on river discharge and flood depth over the past year, specifically in 2014. The model simulation has historically been capable of producing several outputs, such as river discharge and flood depth in Java Island. Based on the result, August is the month with the highest river discharge, which consequently causes the inundation area to be at its maximum condition. When comparing the simulated river network and watershed map on the island of Java, it becomes evident that the CaMa-Flood model has limitations. It cannot represent lakes and reservoirs, and it only captures the main river network, excluding smaller rivers. To overcome this, the study uses Earth2Observe (E2O) data with a 0.25° resolution, which enables the evaluation of CaMa-Flood's capability to reproduce detailed information from diverse runoff data sources. The runoff data used in this study is limited to the years 1980–2014. Therefore, it is necessary to utilize the most recent runoff data (e.g., from 2023). Several sources, such as ERA5, JULES, HiGW-MAT, and others, provide runoff datasets that extend to more recent years. However, since the CaMa-Flood model requires daily-based inputs, it is crucial to select

runoff data available at a daily temporal resolution**.** These findings are highly relevant to climate projections that use low-resolution datasets and are expected to support future research in analysing potential changes to water resources in Java Island under various climate change scenarios.

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